Developing a Home Security System

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A security system is necessary for any location that needs to have its personnel limited or contents protected. Traditional security systems are found in many homes in the United States, and this report will cover the development of a similar system for residential usage. The customer requirements were laid out, and the coding structure was outlined. Ladder logic was used to develop the system through the usage of an HMI, and a PLC. The HMI served as the interface and featured the keypad along with the status screen. The PLC is where the system status was stored and monitored. This home security system features armed and disarmed states, a set amount of time to disarm the system after the door is triggered, a four digit passcode, and a functional panic button when the system is armed or disarmed.

"We certify that all the writing presented here is my own and not acquired from external sources (including generative AI and the student lab manual). We have cited sources appropriately and paraphrased correctly. We have not shared our writing with other students, nor have we acquired any written portion of this document from past or present students."

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I. Introduction

PLC's represent a massive step forward in industrial and residential technological development. Programmable logic controllers (PLCs) use microprocessors and programmable memory to store and implement instructions. These instructions drive multiple functions, including sequencing, counting, timing, and logic, which control processes and machines. PLCs can be easily programmed by uploading a program to its memory. They have the capacity to withstand harsh environments, such as the floor of a factory. The PLC replaced electronic relays before it, which allowed for better control, easier reprogramming, and a smaller form factor.

II. System Requirements

Modern security systems have been developing for years and therefore have become more complex than their predecessors. The system designed today requires a minimum of one sensor input, a keypad to enter a password, a state for arm, warn, and alert. When the door is detected to be opened, a timer must begin where the system is alerted to an intruder but allows time for the alarm to be disabled. This time is set through statistical analysis of travel time and varying input errors. In addition to these system features, the system should be designed with expandability in mind, and allow for the simple addition of new sensors, and the ability to change the password. These features will allow a more modular system which would be more applicable in more environments.

III. System Contents

A. PLC Ladder Logic

Below is our ladder logic code for our PLC system followed by an in-depth explanation of how it works.

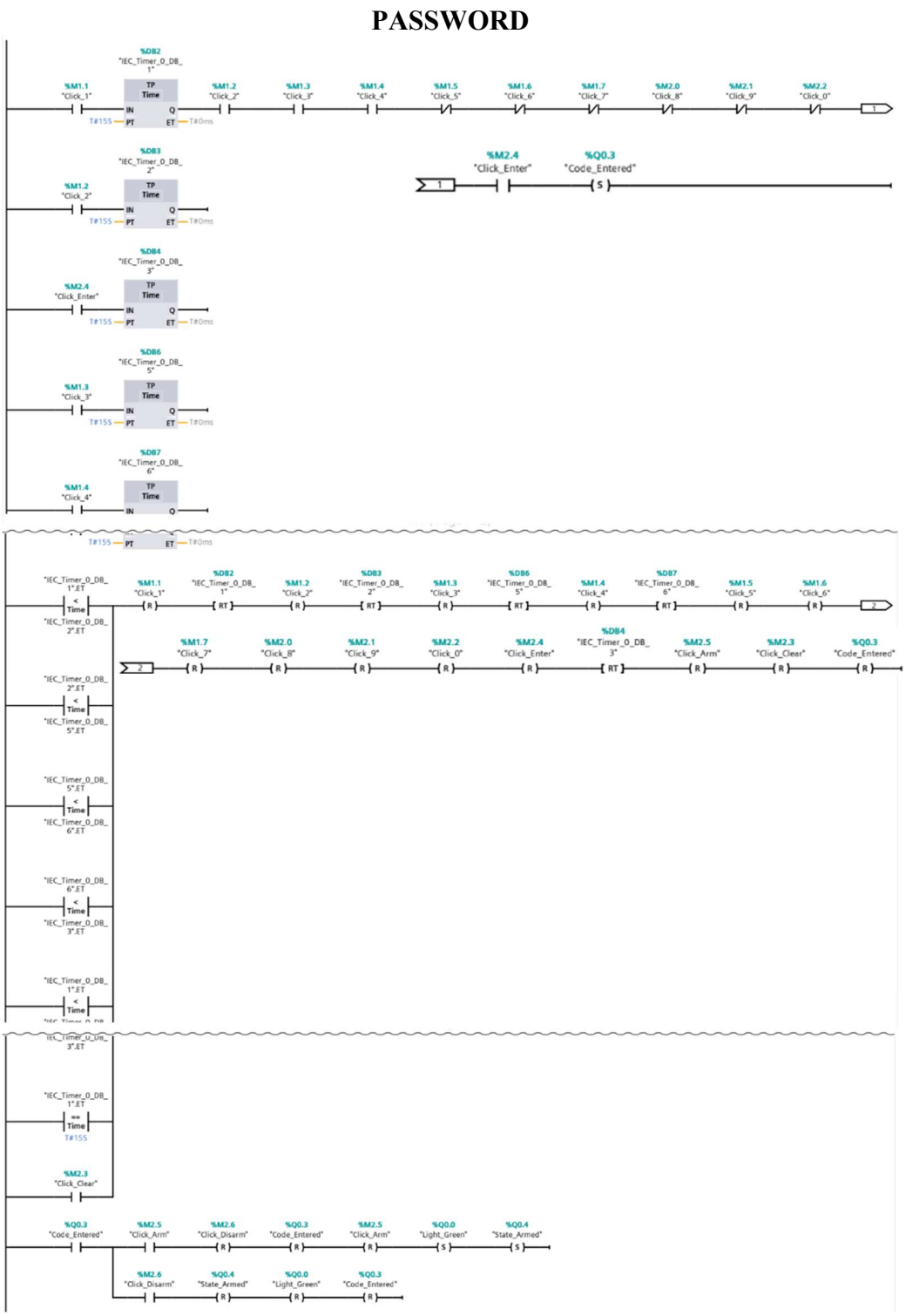


FIG 1 Password PLC Ladder Logic Code for Security System Device

This logic is responsible for setting the code required to arm and disarm the system. To arm the system, the correct series of buttons to press on the HMI is as follows: “1-2-3-4”, “Enter”, “Arm”. To disarm the system one would have the same series of presses but instead of pressing “Arm” one would press “Disarm” instead. In this logic, the top rung sets the code to “1-2-3-4” via the normally open contacts connected to the HMI. If anyone were to press another number instead of those four, the normally closed contacts would open resulting in a failed code. The timers force the buttons to be pressed in the correct series via the comparative rungs, and if the buttons are pressed in the wrong order the comparative operators would read true and immediately reset the code inputs. The same is achieved if one presses the “Clear” button. If the code is entered correctly and the “Enter” button is pressed, the state “Code_Entered” is set to true. This allows the “Arm” or “Disarm” functions to be used as seen in the bottom rung. The logic then either sets “State_Armed” to true or false which is an input in the next ladder logic code, “Main”.

MAIN

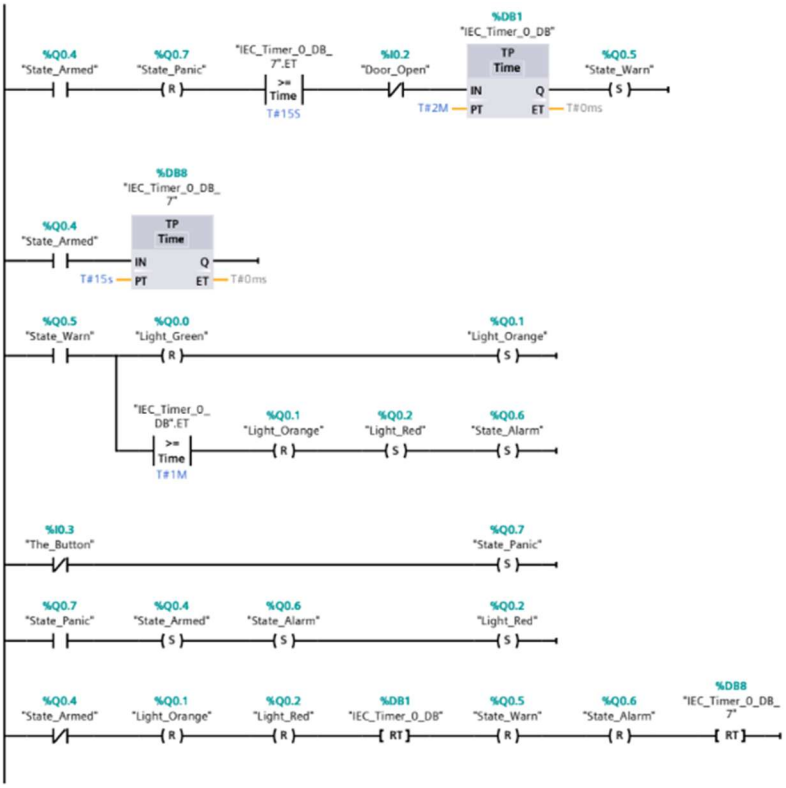


FIG. 2 Main PLC Ladder Logic Code for Security System Device

This logic is responsible for monitoring the sensor while the system is armed, and alerting the user when the sensor is triggered as well as when the alarm is activated. When “State_Armed” is made true by the Password ladder logic, a 15 second timer is started to allow time for the user to leave the premises. After this time, whenever the door is

opened, the signal to “Door_Open” will shut off triggering the warning “State_Warn”, turning off the green light and turning on the yellow/orange light as well as starting a timer. If this timer reaches 1 minute without the disarming sequence being activated, the orange light will turn off and the red light will turn on signaling the alarm has been activated and the authorities have been contacted. At any time during this sequence (whether the alarm has been activated or not) the user may enter the correct code and disarm the system to reset the system to the initial state. However, if the “State_Alarm” is ever true, the authorities will be contacted immediately even if the system is disarmed.

For our above and beyond criteria, a “Panic” button is also introduced that can trigger the alarm at anytime. When the button is pressed, “The_Button” signal turns off triggering “State_Panic” which then arms the system, activates “State_Alarm”, and turns on the red light.

B. HMI Design and User input

The human machine interface (HMI) represents the point of contact between the user and the system. While the user will technically interact with motion and door sensors, this is where the user will interface with the system. It is for this purpose that the display was designed. The root screen for the HMI can be seen below in figure X.

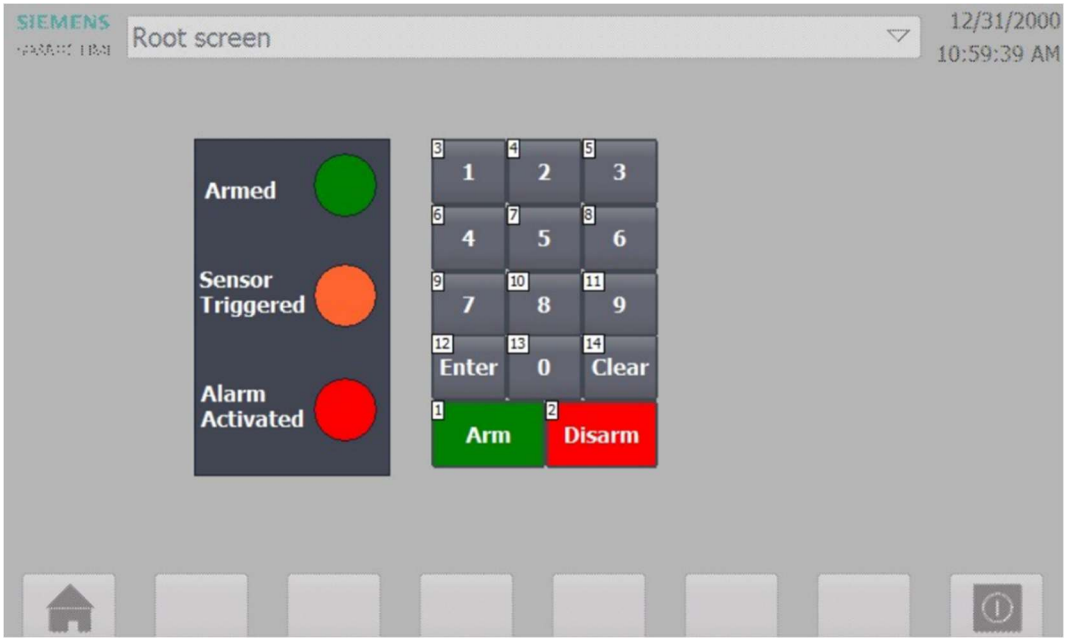


FIG. 3 HMI root screen with status and keypad.

The screen features a three light system intended to display system status at a glance. These states follow the states highlighted in the ladder logic, and are “armed”, “warn” and “alert”. Each of these being tied to a color and easily

identifiable from a distance allows the user to glance at the screen for the status and not have to interact with the system to confirm status.

The keypad was also designed with elements to make usage easier. These elements include uniform and easily spaced buttons, a distinct button for enter and clear, and the arm and disarm buttons being color coded. This combination of design decisions will hopefully reduce the amount of false inputs. HMI design is a balance of small tradeoffs to ensure anyone can understand how to operate the system. With added complexity, there is a loss of simplicity that makes a system more accessible to users. In the context of home security systems, it is a fair assumption that the majority of users want to feel safe in their home, and a simple set of buttons and status lights is the best way to communicate that as simple as possible.

C. Statistical Analysis

After a door is detected open, the system needs to decide what happens next. The system has no way to detect the difference between the resident and an intruder. All it has received is the input of a door being opened. This will be addressed by allowing a fixed amount of time to disable the alarm. To set the fixed amount of time, it is important to recognize the difference between applications of this keypad. Some may be located further away from the door, and some may be right next to it. The time for the alarm disabling needs to be greater than the minimum amount of time for the resident to navigate to the wall and input the correct code, and less than the time it would take an intruder to cause harm.

This time was experimentally measured through three runs of three tests. The tester was instructed to maintain a similar walking pace through all data collections. The first round was a simple keypad entry, with the second and third adding subsequent errors to the keypad entry. This leaves the third test as the worst case scenario where the code was incorrectly entered twice with the third try disarming the system. The experimental data can be seen below in table X.

Test 1 [s]	Test 2 [s]	Test 3 [s]
13.5	18.6	24.67
16.17	17.98	25.64
13.02	18.41	23.36

Table 1 Experimental data from keypad entry time.

The tests show a large variation of time in the data. This is due to the large amount of intentional human error in the experiment. Nobody enters their same house the same way twice, and the data does a good job capturing that. People enter carrying groceries, suitcases, or children. To further analyze the data, the mean times for each test will

be found, as well as the standard deviation of the data. The mean will show the average times taken for each additional mistake, and the standard deviation will show how close together the times are.

	Test 1 [s]	Test 2 [s]	Test 3 [s]
Mean	13.02	18.41	23.36
Standard Deviation	3.41	0.38	3.15

Table 2 Statistical analysis of keypad entry time.

As seen in table 2, the average time taken to enter the code on the first try is 13.02s, with the second at 18.41, and finally the third try at 23.36. These three points can be used to run a linear regression.

Intercept	Slope
7.923	5.170

Table 3 Linear regression results.

The regression has a slope of 5.170, which predicts that on average, there is a 5.170s increase in entry time for each incorrect code. With these results provided, the time the system stays on 'warn' before setting off the alarm was set to one minute. This is higher than what the data suggests, but as mentioned before there will be times when the alarm is not the first place attended after entering the house. This will also allow for flexibility of placement where the same system could be used in a large or small building where a short timer could not have both. This also removes the necessity to reprogram the time based on the house.

IV. Discussion

PLCs are widely used in industrial and technical settings. Their rugged designs make them perfect for factory floors or other unpredictable settings. They are deterministic machines, and based on inputs and outputs, follow their given logic repeatably. which make them extremely well suited for complex systems like security. Instead of connecting multiple systems together, the PLC can host the sensor arrays and interpret codes entered by users to arm, disarm, or trigger the alarm. PLCs differ from computers in how they are programmed. While computers require coding to achieve objectives, users can create programs for PLCs and upload them directly. By standardizing the setup process, the risk of mismatched code and components is eliminated. While running the entire security system through a single PLC can make it much easier to run, the risk of system failure is much higher if the PLC fails. PLCs require constant power, which leaves them vulnerable to power outages. If the system is equipped with a battery of suitable strength, this would eliminate that risk.

The systems available to consumers share some similarities with the system created during the lab. Both have door sensors and a number pad interface. While common systems have physical number pads, the lab system used a digital screen on an HMI. Both systems were set with arming times. Those arming times allow for users to activate their security system and exit their home before the system is armed. Consumer systems usually produce an auditory signal when set the lab system did not. Security systems are programmed to issue an alert if their alarm timers elapse without successfully entering the correct code. While the lab system had no ability to issue that alert. A flashing red light was meant to simulate the issued alert. Some security systems are equipped with motion sensors for increased coverage. The lab security system was not equipped with a motion sensor to avoid higher production difficulties.

V. Conclusion

This lab showed that a PLC is a practical platform for a residential security system. A home security system was developed by using a PLC programmed in ladder logic and paired with an HMI. The system met its design requirements. It arms and disarms with a four-digit passcode entered in a specific sequence; it provides a 15-second exit delay after arming the system. This allows the user to leave the house without triggering the door sensor. The system allows one minute to disarm it after the door sensor is triggered. If the system is not disarmed, the alarm will be activated. There is also a panic button that will trigger the alarm instantly, regardless of the state. The HMI design displays that usability matters as much as the ladder logic underneath it. Three color-coded status lights and a simple keypad with distinct Arm, Disarm, Enter, and Clear buttons allow the user to check the system briefly. Because users depend on a security system to feel safe, simplicity was a deliberate design choice to ensure accessibility. Future improvements to this security system would include a motion sensor, a user-changeable passcode, and a battery backup.

Appendix

Totally Integrated Automation Portal		

ME444 Section 002 Group 1 / PLC_1 [CPU 1214C DC/DC/DC]

PLC tags

	Name	Data type	Address	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Supervision	Comment
	Start_Button	Bool	%M0.0	False	True	True	True		
	Toggle_Switch_1	Bool	%I0.7	False	True	True	True		
	Toggle_Switch_2	Bool	%I0.6	False	True	True	True		
	Toggle_Switch_3	Bool	%I0.5	False	True	True	True		
	Toggle_Switch_4	Bool	%I0.4	False	True	True	True		
	Door_Open	Bool	%I0.2	False	True	True	True		
	The_Button	Bool	%I0.3	False	True	True	True		
	Motion_Sensor	Bool	%I0.0	False	True	True	True		
	Light_Green	Bool	%Q0.0	False	True	True	True		
	Code_Entered	Bool	%Q0.3	False	True	True	True		
	Click_1	Bool	%M1.1	False	True	True	True		
	Click_2	Bool	%M1.2	False	True	True	True		
	Click_3	Bool	%M1.3	False	True	True	True		
	Click_4	Bool	%M1.4	False	True	True	True		
	Click_5	Bool	%M1.5	False	True	True	True		
	Click_6	Bool	%M1.6	False	True	True	True		
	Click_7	Bool	%M1.7	False	True	True	True		
	Click_8	Bool	%M2.0	False	True	True	True		
	Click_9	Bool	%M2.1	False	True	True	True		
	Click_0	Bool	%M2.2	False	True	True	True		
	Click_Clear	Bool	%M2.3	False	True	True	True		
	Click_Enter	Bool	%M2.4	False	True	True	True		
	Click_Arm	Bool	%M2.5	False	True	True	True		
	Click_Disarm	Bool	%M2.6	False	True	True	True		
	State_Armed	Bool	%Q0.4	False	True	True	True		
	State_Warn	Bool	%Q0.5	False	True	True	True		
	Light_Orange	Bool	%Q0.1	False	True	True	True		
	Light_Red	Bool	%Q0.2	False	True	True	True		
	Toggle_Green	Bool	%M2.7	False	True	True	True		
	State_Alarm	Bool	%Q0.6	False	True	True	True		
	State_Panic	Bool	%Q0.7	False	True	True	True		

FIG. 4 PLC tags for all blocks.

References

[1] Wagner, J. R., ME 4440 - Mechanical Engineering Lab III Student Manual, Fall 2026, Department of Mechanical Engineering, Clemson University, Clemson, SC, 2026, pp. 9-12.